Critical and Massive Machine Type Communication towards 6G

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***Abstract*—Critical and Massive Machine-Type Communications (MTC) is one of the key advancements introduced with 5G, but with the more demanding requirements for MTC, emerging use cases and applications towards 2030 will give rise to new and more rigorous requirements on wireless connectivity and MTC. So, the next generation, 6G, will require a new, highly agile and cognitive architecture that can meet these growing requirements. It will be sustainable and "Green" by design, trustworthy and highly secure.**

This white paper explores the main drivers, use cases and requirements of MTC towards 6G, and discusses different types of enabling technologies. We further explore the new Key Performance Indicators (KPIs) of MTC in 6G, followed by key enablers towards energy efficient MTC devices, mission critical MTC and security and privacy for MTC. The paper concludes with the sustainable development goals for 6G MTC.

***Index Terms*—6G, mMTC, IoT, cell-free network, 6G sustainability**

I. Introduction

Over the past 40 years, cellular communications have brought unprecedented benefits to humankind. 2G and 3G unleashed the potential of mobility and connectivity. 4G gave us greater access to information and social engagement. 5G enables the enterprise and extended reality (XR) experiences. By focusing on energy efficiency, automation, infrastructure sharing and efficient spectrum utilization, 6G promises to deliver a network that is not only more advanced in terms of capabilities but also more economical to operate. Every mobile network generation yields new features and a wealth of new opportunities. A significant aspect of this evolution has been the continuous advancement of Machine-Type Communication (MTC). Early attempts in 2G and 3G laid the groundwork for device connectivity, primarily through Machine-to-Machine (M2M) communication and basic Internet of Things (IoT) implementations. 4G introduced more robust IoT capabilities with technologies such as Narrowband IoT (NB-IoT) and LTE-M (Long-Term Evolution Machine Type Communication), allowing for greater device density, extended battery life, and broader application support.

Now as 5G is rolling out commercially, enabling a rich fabric of cutting-edge technologies powering smart cities, factories, precision farming, the Internet of Things (IoT) and robotics, the rising demand and strain on the network will require a new, highly agile and cognitive architecture that automatically deploys new services optimally tailored to these applications, while further pushing the limits for the support of ultra-low latency, massive capacity and widespread connectivity. Not to mention better energy efficiency. 6G will meet these expectations and even go beyond by reformulating the very concept of network communications. While 5G has already laid the groundwork for massive Machine-Type Communication (mMTC), 6G is expected to take these capabilities to new heights, especially in terms of reliability, latency, scalability, and intelligence. These topics are covered in this paper, which is mostly based on [1].

II. MTC Megatrends

A. Key drivers and use cases

The evolution of MTC in the 6G era will be influenced by key drivers across various sectors [3]. Autonomous mobility will see autonomous vehicles, including self-driving cars, AGVs, and UAVs performing coordinated tasks, becoming mainstream by 2030. This will increase the demand for advanced, 3D network connectivity due to the rise in sensors and automation levels.

In connected living, widespread IoT device deployment will connect everything by 2030, enhancing wearables and integrating new devices like smart clothing and bio-implants, fostering smart living and smart city infrastructure.

Industry 5.0 will build on Industry 4.0 by focusing on personalized production with interactive environments involving humans and machines, supported by extensive connectivity and agility.

Advancements in full digital immersion, such as brain-computer interfaces and XR, will enable the Internet of Senses, allowing immersive interactions and control in remote environments.

Towards 'zero-energy', energy-harvesting devices with ultra-low power design will ensure perpetual operation, aligning device service life with product life to enhance sustainability.

Finally, as data becomes the new oil, MTC networks will efficiently mine and utilize data through AI and DLT, expanding applications for authenticated data exchange and services by 2030.

B. Requirements and service classes

In 5G, MTC is mainly divided into URLLC and mMTC. URLLC focuses on controlled settings with small data loads and limited data rates, while mMTC handles large, dense deployments with intermittent traffic from many devices. Moving toward 6G, MTC requirements become more varied and

complex, driven by new use cases and specialized service needs.

Compared to 5G, 6G QoS requirements will be more stringent, necessitating ultra-dense deployment of UAV swarms and industrial IoT devices with 3D connectivity supporting up to 10 connections per m^3. Applications like the Internet of Senses and Industry 5.0 will require higher data rates with enhanced reliability and latency, targeting peak spectral efficiency around 40 bpcu, an increase from 25 bpcu in 5G NR (max download speed of 2.5 Gbps over 100 MHz).

E2E KPIs are essential for real-time optimization and collaboration in industrial settings. Closed-loop control applications will need E2E reliability of up to 99.999% with latencies below 1 ms, implying per-link reliability of around 99.9999% and user plane latency of 0.1 ms.

Energy efficiency is crucial, focusing on the total cost and energy consumption per delivered bit, including the environmental impact. The goal for 6G is Zero Energy Mobile Terminals (ZE MTDs), achieved through efficient Ultra-Low Power (ULP) hardware (with power consumption under 1 nanowatt in standby and 1 microwatt active) and energy harvesting techniques.

New KPIs will include Age of Information (AoI), measuring information freshness vital for monitoring systems, and interoperability across heterogeneous networks. Dependability will integrate attributes like availability and reliability, characterizing system life cycles. Localization accuracy, increasingly important in 6G, will enhance positioning services. Another key KPI is connection set-up time, defined as the duration from packet arrival at a transmitter to readiness for transmission.

A summary of the prospective MTC KPIs in 6G and their comparison with 5G KPIs is presented in the Table I [3].

Based the requirements and use cases, URLLC and mMTC from 5G NR will evolve into distinct service classes.

Dependable cMTC will provide extreme E2E reliability, low latency, security, and precise localization for applications like autonomous mobility.

Broadband cMTC will support high-reliability MBB data with low latency for multi-sensory XR, digital twin-controlled factories, and robotic surgery.

Scalable cMTC combines URLLC and mMTC, offering massive connectivity with moderate reliability and latency for body area networks, smart city monitoring, and factory automation.

Globally-scalable mMTC will provide ultra-wide coverage, including volumetric density of devices, enabled by UAV swarms and NTN for global connectivity.

ZE mMTC aims for energy-efficient, perpetual devices for applications like soil monitoring in precision agriculture, utilizing ZE radios, ULP hardware, energy harvesting, backscatter communication, and extreme EE resource allocation.

TABLE I
A COMPARISON OF SELECTED MTC KPIS IN 5G AND 6G

KPI	5G target	6G target
Per radio link reliability	$1 - 10^{-5}$	$1 - 10^{-9}$
Application level E2E reliability	Not considered	$1 - 10^{-7}$
Per radio link latency	1 ms	0.1 ms
Application level E2E latency	5 ms	$<$ 1 ms
Connection set-up time	Not considered	$<$ 1 ms
Connection density	1 device/m^2	Up to 10 device/m^3
Spectral efficiency (downlink)	$\sim$ 25 bpcu	$\sim$ 40 bpcu
Device lifetime	10 years	40 years
Energy consumption	Low	Ultra-low
Positioning accuracy	30 cm	10 cm
Jitter	1 μs	$<$ 0.1 μs
E2E optimization	Not considered	Relevant
Dependability	Not considered	Relevant

III. ENABLING TECHNOLOGIES FOR 6G MTC

A. MTC Network Architecture

In the progressive domain of Machine-Type Communication (MTC), the network architecture is expected to experience a shift towards more fluid, cell-free paradigm from the traditional cell-based network architecture. In this new model, the rigid connections between Machine-Type Devices (MTD) and specific base stations (BS) are dissolved, which leads to reducing the signal overhead and concurrently improving the reliability of the transmissions. This is achieved through the collaborative efforts of multiple base stations, which either independently or collectively demodulate incoming transmissions.

Moreover, the network landscape is experiencing an escalation in heterogeneity. This is due to several factors: the emergence of local micro-operators who are pioneering new private or public networks, realising the 'crowd-driven networking' concept at one end of the spectrum; the introduction of novel Radio Access Technologies (RATs) alongside the refinement of existing ones; and the reduction of cell sizes. This latter development is driven by the migration to higher frequency bands and the adoption of advanced modulation and coding techniques, leading into highly specialized deployments that are geographically concentrated and tailored to meet specific objectives. This heterogeneity escalation and the reduction in machine-BS inter-connectivity is visualised in figure 1.

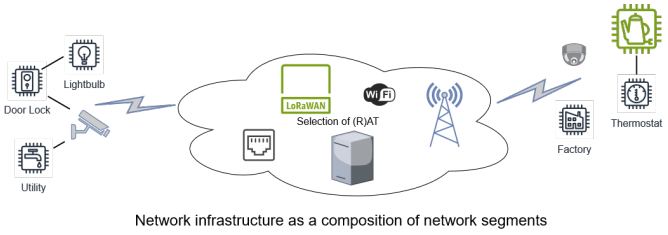


Fig. 1. Visualisation of the network heterogeneity

For enhanced efficiency, the network must inherently support a dynamic and cooperative orchestration among end-to-end (E2E) applications. These applications are able to use various network domains, such as wired, optical, and wireless, while ensuring seamless interoperability. The interoperability requires technology-neutral interfaces for the different access technologies. This means that while end-to-end applications

may not need to understand the specific underlying technologies and Radio Access Technologies (RATs), they should nonetheless be capable of conveying their network requirements effectively. The choice of Radio Access Technologies (RATs) should be determined by transparent algorithms that consider the E2E Quality of Service (QoS) demands of different applications [1].

It is important to recognise that the availability of RATs, quantity and arrangement of machine-clients, as well as their traffic behaviors and communication requirements, are dynamic, as they can undergo significant alterations across different locations and over time. This implies a need for real-time localisation and optimisation processes that can be deployed to the network edge, as the processes can significantly increase the network performance [4]. Additionally, an introduction of emergency traffic channels would be appreciated [1].

B. Energy Efficient MTC Devices

With billions of devices expected as part of massive MTD deployments in 6G, energy efficiency is set to be an important KPI, especially for long-term sustainability. Improved hardware design in MTDs can greatly boost energy efficiency through technologies such as ultra-low-power (ULP) transceiver architecture, advanced energy harvesting (EH) methods, and ambient backscatter communication (AmBC).

ULP transceivers achieve energy savings by using smaller radiating elements, which result in narrower RF bandwidths and better frequency selectivity. Power consumption can be further reduced by using event-driven architecture, which adjusts power use by switching between modes based on need. From the transmission side, creating efficient waveforms and maintaining linearity in amplification are crucial to maximizing energy and avoiding power loss. The choice of transmission band and modulation scheme also affects power use and sensitivity; for instance, sub-GHz bands combined with simple, non-coherent modulation like On-Off Keying (OOK) can achieve high energy savings with good sensitivity [5].

Enhanced energy harvesting allows the device to stay in a low-power sleep mode, becoming active only when it receives a RF wake-up signal (WUS). The WUS preamble supplies enough energy to convert the signal into power, allowing the device to decode and process the information in the frame [6]. However, the device must be near the RF source and requires complex hardware for EH to function. In the future, these improvements could support battery-free operation in some MTDs, with WUS signals powering downlink and AmBC supporting uplink communication.

AmBC allows devices to communicate by modulating and reflecting existing RF signals, rather than emitting their own signal through waveform generation and amplification. This allows AmBC devices to operate with much lower power consumption, as the reflected signals appear as additional multi-path components to the receiver. While promising, this approach also faces challenges, particularly in separating interference from useful reflected signals and handling the higher path loss associated with reflected AmBC signals.

Together, these advancements in ULP architecture, enhanced energy harvesting, and ambient backscatter communication represent promising steps toward realizing ultra-low-power MTDs, helping to enable sustainable and energy-efficient MTC in future 6G networks.

C. Global Massively Scalable MTC

Driven by the growing demand for smart city technologies such as autonomous mobility, intelligent transportation, and industrial automation, one of the key emerging service categories in 6G is globally scalable massive Machine-Type Communication (mMTC). This service class is designed to support large-scale coverage and networks comprising an extensive number of devices [1]. It emphasizes connection density and energy efficiency, while imposing less stringent requirements for reliability, latency, and data rates.

First and foremost, a critical research question in 6G mMTC is how to enable disruptive design for 6G that addresses challenges like minimizing power consumption, mitigating interference in ultra-dense deployments, and ensuring interoperability in heterogeneous networks. Frequency spectrum regulation is a crucial enabler, and since most of the high frequency bands that will be used in 6G are unregulated, this is a suitable opportunity to achieve global spectrum harmonization. However, it should be noted that this requires strong political collaboration between different countries' regulatory bodies [1].

In terms of the technological aspect, while 5G incorporates numerous advanced technologies for MTC, the inherent differences between MTC and Human-Type Communication (HTC) like traffic direction density render 5G fundamentally inefficient for addressing mMTC requirements. Consequently, new research directions are emerging across various layers of the communication stack, from the physical (PHY) layer to the application (APP) layer. At the physical layer, several critical areas of investigation are noteworthy. To address the scalability demands of massive traffic, non-orthogonal multiple access (NOMA) schemes are employed to mitigate resource collisions. For this use case, advanced NOMA systems work with GF transmission, so low-complexity multi-user detection (MUD) algorithms require careful design considerations for success. Another crucial aspect is the adoption of error-correcting coding schemes. Given the constraints of mMTC's medium access control (MAC) protocols, 6G necessitates enhanced error detection capabilities, channel state information (CSI)-free decoding algorithms, and diverse coding schemes. This highlights the need to develop novel types of codes tailored to different service classes and control information. At the MAC layer, grant-free (GF) transmission is pivotal, as it eliminates the control signaling overhead and facilitates traffic differentiation based on quality-of-service (QoS) [7], which is particularly advantageous for periodic, time-sensitive traffic. On the other hand, mMTC also involves unpredictable traffic patterns, necessitating modern random access schemes. This

remains an active area of research, with proposed solutions including mechanisms for multi-user code construction and joint deconstruction. It should be noted that one of the enabling technologies for 6G is the deployment of small cells with massive receiver numbers, which has the potential to reduce the design burden associated with random access schemes. In general, mMTC exhibits various traffic and QoS characteristics (sporadic, periodic, event-driven traffic...), so network slicing technology would be beneficial in supporting different needs by different schemes (random, persistent, hybrid...). On the other hand, a critical enabler for global mMTC is NTNs technology. Advancements in this area would significantly offload traffic from terrestrial networks and promote remote area cellular connection. While commercial solutions in LEO are actively being developed, the issue of Doppler effect and efficiency in information routing calls for new physical waveform and network architecture design. Research areas on the higher layers are implicit, with noticeable focus being on the ability to merge interactive procedures in the application layer where there is no connection state information as an effect of GF transmission protocol. Global mMTC also means the importance of orchestrated massive software update is high, but devices spending energy on monitoring update announcement is inefficient in this use case. Spending research resource on mechanism like on-demand paging is therefore important [1].

D. Mission Critical MTC

Mission Critical MTC (cMTC) is one of the primary focuses of MTC in 6G aiming at providing even better reliability and latency as URLLC in 5G. The goal is to have as low latency as 0.1 ms and block error rate (BLER) of 10^{-9} [1]. This would mean that different verticals such as industrial automation could leverage 6G for connecting critical machinery wirelessly for the same reliability as with cables [1]. To make cMTC communications scalable, there needs to be some modifications on the initial idea of URLLC in 5G [3].

6G needs to introduce new service classes that are tailored for specific communication scenarios. These classes should take into consideration the specific application needs and the predictability of resource requirements and conditions [3]. Since not all critical applications require sub-millisecond end-to-end (E2E) latencies, the focus should be on providing resource-efficient solutions with guaranteed latency and BLER specified for the application [3].

Most machine-type communications can be very deterministic such as periodic control and status messages from a machine operating in a smart factory. This communication can be efficiently scheduled under the specific time constraints [1]. More issues arise when dealing with events and alarms in which the inter-arrival times (IAT) are not known. Events can be handled by allocating resources but that can lead to inefficiency as the reserved resources might be unused. Thus, the 6G cMTC service classes should also take into consideration the IAT distribution and try to predict it with the help of machine learning [1].

In order for a 6G system to be able to make decisions involving different service classes, a specific cMTC management function needs to be implemented [1]. This function is responsible for the resource monitoring and prediction of multiple networks connected to it. Resource awareness is one of the key enablers of cMTC, and it also requires that different networks inform cMTC management function about their load level and other parameters.

One of the most important building blocks of cMTC in 6G is radio resource management. This management can be costly for power-constrained end-devices. Thus, resource management and decision process can be done by a broker function inside the cMTC management function to conserve the energy of end-devices. The broker function could manage the scheduling and spectrum resources across different radio-access technologies (RAT). It can also operate a digital twin of the cMTC device and use it to simulate and evaluate decisions with the help of machine learning algorithms [1].

Although cMTC aims to minimize collisions by resource allocation, they can not be fully avoided. Collisions will happen when the network topology varies rapidly for instance in vehicle-to-vehicle (V2V) communications, where transmissions are very frequent and mobility is high [1]. Thus, cMTC devices should use a transceiver able to provide reliable communication despite collisions. These transeivers could use non-orthogonal multiple access methods (NOMA) to separate signals in code, power or spatial domain to avoid random non-orthogonality in grant-free multiple-access channels. Furthermore, full-duplex communication needs to be ensured for simultaneous transmission and receiving [3].

E. Privacy and Security for MTC

Privacy and security have always been major concerns for any system, especially in sensitive areas. Even today, these issues remain critical. Looking to the future, such as the development of 6G technology, privacy and security will continue to be significant challenges. The strength of a security system relies on its weakest point. Hackers only need to exploit a single vulnerability to compromise security, which is why building a perfect security system is extremely challenging. Although MTC (Machine-Type Communication) does not directly involve humans, it still faces certain challenges. These include: i) Autonomous Mobility ii) Connected Living iii) Factories of the Future. [1] However, these problems can be mitigated using advanced technologies, such as OAuth 2.0. Additionally, Distributed Ledger Technology (DLT) and smart technologies may provide more suitable solutions, especially in the context of 6G Machine-Type Communication (MTC). Although these technologies are powerful, they still need to overcome certain challenges in some IoT device use cases, particularly in two-way communication and resource management.

When discussing privacy and security, we must also consider cost and scalability. Most conventional methods are not effective when dealing with a large number of devices. To address these challenges, we can adopt new authentication techniques,

such as group-based authentication or physical layer authentication. Additionally, machine learning techniques, like utilizing Physical Unclonable Functions (PUF), can provide effective solutions. PUF is a type of hardware-based security solution. Examples of PUF applications include unique fingerprinting, key generation, and on-the-fly key generation.
Security is never perfect; it evolves by addressing vulnerabilities. It is a continuous process. Detecting anonymous entities is crucial in some cases, but current tools have certain limitations. However, technologies and security tools, such as PUFs mentioned earlier, continue to advance.

IV. Sustainable Development Goals for 6G MTC

The United Nations Sustainable Development Goals (UN SDGs) represent a global framework for achieving a better and more sustainable future for all. Adopted in 2015, the 17 goals aim to address pressing global challenges, including poverty, inequality, climate change, and environmental degradation, by 2030. [2] It is important for communication technologies, like 6G to support these objectives.

Sustainability's importance is evident as it has been established as a new key performance indicator (KPI) for 6G, which must be taken seriously. Its impact can be seen, for example, in the development of energy-efficient devices. The impact of MTC on sustainability is significant, as MTC is designed to address various energy efficiency and resource management challenges within 6G. [1] MTC networks aim to support sustainable growth by enabling the large-scale deployment of low-power, long-lasting devices that can communicate with minimal energy consumption. This feature can be particularly useful for applications in smart cities, agriculture, and industrial automation, where a large number of devices are deployed, often in remote or challenging environments. [1]

One of the primary sustainability challenges in 6G MTC is to make the network sustainable while maximizing the performance of individual machines, the network as a whole, and the plethora of services and applications served by the network. [1] Achieving sustainability in MTC involves minimizing power usage in devices and managing the carbon footprint effectively. The ability to monitor devices and systems in real-time can help extend the life cycle of equipment by enabling predictive maintenance. Predictive maintenance can prevent unexpected failures and reduce the need for costly repairs, leading to more efficient use of resources. MTC can enhance sustainability also by enabling efficient data collection through machines, and supporting smart, energy-aware applications in industries and urban environments. Proposed solutions in MTC like ULP include designing receivers that use very little power and implementing smart power management methods [1]. These strategies are intended to reduce energy usage when devices are not actively transmitting data.

Wireless energy transfer (WET) aims to provide energy wirelessly. [1] This would reduce battery waste and enhancing device longevity. WET could also enable more reliable device operation in remote or hard-to-reach areas, reducing the need for frequent battery replacements and maintenance. This contributes to sustainability by minimizing both logistical efforts and material waste.

Sustainability is an integral part of the 6G vision, particularly within Critical and Massive MTC, with a strong focus on energy efficiency and resource management. By addressing the energy challenges and advancing efficient resource management strategies, 6G will play a pivotal role in building a more sustainable and connected world.

V. Conclusion

As we transition from 5G to 6G, the demands for both critical and massive Machine Type Communication will continue to grow. 6G will need to address the challenges of ultra-reliable, low-latency communication for critical and massive scalability for massive MTC while ensuring efficiency, security, and sustainability. The successful development of 6G will unlock a host of new possibilities across multiple industries, ranging from healthcare and transportation to smart cities and industrial automation, enabling a fully connected, intelligent, and autonomous future.

References

[1] Mahmood, N. H., López, O., Park, O.-S., Moerman, I., Mikhaylov, K., Mercier, E., Munari, A., Clazzer, F., Böcker, S., Bartz, H. (2020). (Eds.). White Paper on Critical and Massive Machine Type Communication Towards 6G [White paper]. (6G Research Visions, No. 11). University of Oulu. http://urn.fi/urn:isbn:9789526226781

[2] Matinmikko-Blue, M., Aalto, S., Asghar, M. I., Berndt, H., Chen, Y., Dixit, S., Jurva, R., Karppinen, P., Kekkonen, M., Kinnula, M., Kostakos, P., Lindberg, J., Mutafungwa, E., Ojutkangas, K., Rossi, E., Yrjölä, S., Öörni, A. (Eds.). (2020). White Paper on 6G Drivers and the UN SDGs [White paper]. (6G Research Visions, No. 2). University of Oulu. http://urn.fi/urn:isbn:9789526226699

[3] Mahmood, N.H., Böcker, S., Moerman, I. et al. Machine type communications: key drivers and enablers towards the 6G era. J Wireless Com Network 2021, 134 (2021). https://doi.org/10.1186/s13638-021-02010-5

[4] Peltonen, E., Bennis, M., Capobianco, M., Debbah, M., Ding, A., Gil-Castiñeira, F., Jurmu, M., Karvonen, T., Kelanti, M., Kliks, A., Leppänen, T., Lovén, L., Mikkonen, T., Rao, A., Samarakoon, S., Seppänen, K., Sroka, P., Tarkoma, S., & Yang, T. (2020). 6G White Paper on Edge Intelligence [White paper]. (6G Research Visions, No. 8). University of Oulu. http://urn.fi/urn:isbn:9789526226774

[5] David D. Wentzloff, "Low Power Radio Survey," [Online]. www.eecs.umich.edu/wics/low_power_radio_survey.html

[6] Bello, H.; Xiaoping, Z.; Nordin, R.; Xin, J. Advances and Opportunities in Passive Wake-Up Radios with Wireless Energy Harvesting for the Internet of Things Applications. Sensors 2019, 19, 3078. https://doi.org/10.3390/s19143078

[7] G. Karadag, R. Gul, Y. Sadi, and S. Coleri, "QoS-con-strained semi-persistent scheduling of machine type communications in cellular networks," IEEE Transactions on Wireless Communications, vol. 18, no. 5, pp.2737–2750, 2019.